








Time Series Analysis

Seminar 13: Speculative bubbles: LPPL models



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **BET**: Bucharest Exchange Trading (index)
- **BSADF**: Backward Supremum Augmented Dickey–Fuller (statistic)
- **EODHD**: EOD Historical Data (market-data provider)
- **GSADF**: Generalised Supremum Augmented Dickey–Fuller (test)
- **LPPL**: Log-Periodic Power Law
- **OLS**: Ordinary Least Squares
- **PSY**: Phillips–Shi–Yu (tests)
- **SADF**: Supremum Augmented Dickey–Fuller (test)

Today's Question and Seminar Structure

- The question of the seminar
  - **can we see a bubble while it grows, date it, and say when it ends?**
- Results to obtain (practice only: nothing is handed in or graded)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C1
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Chapter 13 is a self-study chapter: read it before the seminar; the recap gives the formulas used today
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	the right-tailed ADF on one window; reading SADF and GSADF	Solved, Proposed	A1
A3	dating a bubble from the BSADF sequence	Solved	A3
A4	the LPPL formula and its filter conditions	Proposed	A3
A5	the recursion of a rational bubble	Solved	A5
B1, B2	the S&P 500 in the dot-com years; LPPL fits of Bitcoin in 2021	Solved, Proposed	B1
C1	what is wrong in an AI answer?	Proposed	A1–B2

- Types: **A** calculations on paper; **B** real data, in the notebook; **C** an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency	Period
S&P 500	EODHD (GSPC.INDX)	weekly, Friday close	1990–2004
Bitcoin	EODHD (BTC-USD.CC)	daily, 7 days a week	2020–2021
Nasdaq 100, Bitcoin, BET, Shanghai	Chapter 13 results	weekly	1990–2023

- Weekly log price $y_t = \ln P_t$ for the explosive-root tests; daily log price for the LPPL fits; time in decimal years
- In the notebook: `adf_window`, `psy`, `psy_cv`, `episodes_above`, `lppl_fit`, `lppl_conditions` (the functions of Chapter 13); no account or key is needed

Recap (1/3): explosive roots and the PSY tests

- **Right-tailed ADF** on a window $[r_1, r_2]$ of the sample (Augmented Dickey–Fuller, no lagged differences here) Phillips, Wu and Yu (2011)

$$\Delta y_t = a + \delta\, y_{t-1} + \varepsilon_t, \qquad H_0 : \delta = 0 \text{ (unit root)}, \quad H_1 : \delta > 0 \text{ (explosive, } \rho = 1 + \delta > 1)$$

  - statistic $\mathrm{ADF}_{r_1}^{r_2} = \hat{\delta}/\mathrm{se}(\hat{\delta})$; reject H_0 when it is **above** the right-tail critical value

- **Sup statistics** (PSY: Phillips, Shi and Yu) Phillips, Shi and Yu (2015a); smallest window $r_0 = 0.01 + 1.8/\sqrt{T}$

$$\mathrm{SADF} = \sup_{r_2 \in [r_0, 1]} \mathrm{ADF}_0^{r_2}, \qquad \mathrm{GSADF} = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}, \qquad \mathrm{BSADF}_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}$$

  - critical values by Monte Carlo under a random walk with a weak drift (1000 replications in the course)

- **Date-stamping**: an episode starts at the first week with BSADF_{r_2} above its 95% critical value and must last at least $L = [\ln T]$ weeks (rounded)

Recap (2/3): rational bubbles

- **Price = fundamental value + bubble**: with a required return r, $p_t = p_t^* + b_t$ and Blanchard and Watson (1982)

$$\mathbb{E}_t(b_{t+1}) = (1+r)\, b_t$$

  - constant dividend D: $p^* = D/r$; a bubble is the part of the price that dividends do not explain
- **Blanchard–Watson**: each period the bubble survives with probability π, otherwise it bursts

$$b_{t+1} = \begin{cases} \dfrac{1+r}{\pi}\, b_t + \varepsilon_{t+1} & \text{with probability } \pi \\ \varepsilon_{t+1} & \text{with probability } 1-\pi \end{cases}$$

  - while it survives the bubble grows faster than r: the extra growth pays for the risk of the crash
  - the duration is geometric: mean $1/(1-\pi)$ periods (as the regimes of Chapter 10)

Recap (3/3): the LPPL model and its filter

- **LPPL** (log-periodic power law) Johansen, Ledoit and Sornette (2000): the log price before a critical time t_c

$$\ln p(t) = A + B\,(t_c - t)^m + C\,(t_c - t)^m \cos\left(\omega \ln(t_c - t) - \phi\right), \qquad t < t_c$$

  - $B < 0$, $0 < m < 1$: faster than exponential growth; ω: frequency of the oscillations in $\ln(t_c - t)$; successive peaks shrink by $\lambda = e^{2\pi/\omega}$
  - Filimonov–Sornette: A, B, C_1, C_2 by OLS (ordinary least squares) for given (t_c, m, ω) Filimonov and Sornette (2013)
- **Filter** of a fit on $[t_1, t_2]$ Shu and Zhu (2020): $B < 0$; $0.01 \le m \le 0.99$; $2 \le \omega \le 25$; $t_2 \le t_c \le t_2 + (t_2 - t_1)/5$

$$\text{oscillations} = \frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \ge 2.5, \qquad \text{damping} = \frac{m\,|B|}{\omega\,|C|} \ge 1, \qquad \text{relative error} \le 0.15$$

  - $C = \sqrt{C_1^2 + C_2^2}$; damping ≥ 1 keeps the fitted price increasing; t_c is the most likely end of the regime, not a crash date

A1: the right-tailed ADF on one window [Solved]

Task

- Weekly S&P 500 log prices, 5 January 1990 to 24 March 2000 (the peak week): $n = 533$ weekly changes. OLS gives $\hat{a} = -0.0084$, $\hat{\delta} = 0.00175$, $\text{se}(\hat{\delta}) = 0.00180$.
- 1. Compute the ADF statistic and the root $\hat{\rho}$.
- 2. The 95% critical value of BSADF for this week is 0.542. Is the week explosive?
- 3. The weeks above the critical value around the peak run from 3 March 2000 to 7 April 2000; with $L = 7$, is this a dated episode?
- Report: one statistic, one root, two decisions.

Solution

- 1. ADF $= 0.00175/0.00180 = 0.970$; $\hat{\rho} = 1 + \hat{\delta} = 1.00175$
- 2. $0.970 > 0.542$: reject the unit root in favour of an explosive root (right tail); a left-tailed reading ($t < -2.86$) would wrongly accept
- $\hat{\rho}$ is barely above 1: a deviation doubles in $\ln 2 / \ln \hat{\rho} \approx 398$ weeks; mild explosiveness is enough to be detected
- 3. No: the run lasts 6 weeks, fewer than $L = 7$; the minimum duration removes short crossings

TSA_ch13_seminar

Back Exercise list

A2: the data [Proposed]

Series (weekly)	T	w_0	SADF	cv_{95}	GSADF	cv_{95}	Episodes
Nasdaq 100, 1990–2004	782	?	2.722	1.481	2.925	2.276	6
Bitcoin, 2014–2023	484	?	3.849	1.495	3.849	2.184	2
BET, 2000–2012	663	52	1.859	1.485	3.533	2.294	3
Shanghai, 2010–2018	460	43	1.501	1.406	3.458	2.147	1

- Results of Chapter 13 (psy, psy_cv, 1000 replications); cv_{95}: 95% critical value; w_0: smallest window in weeks
- Nasdaq 100, whole-sample ADF: -1.272, with the 95% critical value -0.132

TSA_ch13_seminar

A2: reading SADF and GSADF [Proposed]

Solution discussed in class

Task

- The table of the previous slide. Model: A1.
- 1. Compute $r_0 = 0.01 + 1.8/\sqrt{T}$ and $w_0 = \lfloor r_0 T \rfloor$ for the Nasdaq 100 and for Bitcoin.
- 2. For each series, decide on SADF and on GSADF at 5%.
- 3. Why is GSADF never below SADF? Why are they equal for Bitcoin?
- 4. Why does the whole-sample ADF of the Nasdaq 100 not reject?
- Report: two windows, eight decisions, two sentences.

A3: the data [Solved]

Week	BSADF	cv_{95}
26 May 1995	0.188	0.484
2 June 1995	0.518	0.427
9 June 1995	0.219	0.436
16 June 1995	0.657	0.443
23 June 1995	1.022	0.444
30 June 1995	0.668	0.473

Week	BSADF	cv_{95}
7 July 1995	1.090	0.461
14 July 1995	1.149	0.419
21 July 1995	0.745	0.475
28 July 1995	1.046	0.475
4 August 1995	0.783	0.475
11 August 1995	0.563	0.433

- Weekly S&P 500, 1990–2004 ($T = 782$ weekly changes, B1); BSADF and its 95% critical value, weeks around the first crossing that lasts

TSA_ch13_seminar

A3: dating a bubble from BSADF [Solved]

Task

- The table of the previous slide; the episode rule of the recap.
- 1. Compute the minimum duration $L = [\ln T]$.
- 2. Mark the weeks with BSADF above the critical value.
- 3. Give the start of the first episode. Why is the first crossing not the start?
- Report: L, a start date, one sentence.

Solution

- 1. $\ln 782 = 6.66$, so $L = 7$ weeks
- 2. Above: 2 June 1995 ($0.518 > 0.427$), then every week from 16 June 1995 on
- 3. The episode starts on 16 June 1995 and lasts 57 weeks (to 12 July 1996); the crossing of 2 June 1995 lasts one week, fewer than L
- Over the whole sample, 8 of the 13 crossings are shorter than L and are dropped

TSA_ch13_seminar

A4: the LPPL formula and its filter [Proposed]

Solution discussed in class

Task

- LPPL with $A = 8$, $B = -1$, $C = 0.1$, $m = 0.4$, $\omega = 7$, $\phi = 0$, $t_c = 1$ (time in years). Model: A3.
- 1. Compute $\ln p(t)$ at $t = 0,\ 0.5,\ 0.9$, term by term.
- 2. Compute $\lambda = e^{2\pi/\omega}$ and the first three peaks of the cosine after $t = 0$.
- 3. For a window $[t_1, t_2] = [0, 0.9]$, compute the oscillations and the damping. Which filter condition fails?
- Report: three log prices, λ, three dates, two numbers, one verdict.

A5: the recursion of a rational bubble [Solved]

Task

- Blanchard–Watson bubble with $r = 0.01$ per period, $\pi = 0.95$, $b_0 = 1$, no noise; a constant dividend $D = 2$.
- 1. Compute the growth factor while the bubble survives and the path b_1, b_2, b_3.
- 2. Check that $\mathbb{E}_0(b_1) = (1 + r)\, b_0$.
- 3. Compute the probability of surviving three periods, the mean duration and the median duration.
- 4. Compute p^* and the share of the bubble in the price after three periods.
- Report: one factor, three values, three probabilities or durations, one share.

Solution

- 1. $(1 + r)/\pi = 1.01/0.95 = 1.0632$ (+6.32% per period); $b_1 = 1.0632$, $b_2 = 1.1303$, $b_3 = 1.2017$
- 2. $\mathbb{E}_0(b_1) = \pi \cdot 1.0632 \cdot 1 + (1 - \pi) \cdot 0 = 1.01 = (1 + r)\, b_0$
- 3. $\pi^3 = 0.857$; mean $1/(1 - \pi) = 20$ periods; median $\ln 0.5 / \ln \pi = 13.5$ periods
- 4. $p^* = 2/0.01 = 200$; $p_3 = 201.20$, bubble share 0.60%: a young bubble is invisible in the price; it matters only after many periods of survival

TSA_ch13_seminar

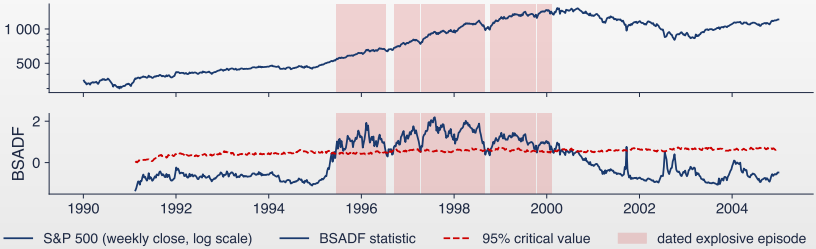
Back Exercise list

B1: the S&P 500 in the dot-com years [Solved]

- **Question**: was the S&P 500 explosive in the late 1990s, and when, compared with the Nasdaq 100 of Chapter 13?
- **Inputs**: weekly S&P 500 log prices, January 1990 to December 2004 ($T = 782$), the same sample as the Nasdaq 100 of the lecture
- **Tasks**
  1. Compute ADF, SADF, GSADF and the BSADF sequence (`psy`) and their critical values (`psy_cv`, 1000 replications).
  2. Date the episodes with the 95% critical value of BSADF and $L = [\ln T]$ (`episodes_above`).
  3. Compare the statistics and the longest episode with those of the Nasdaq 100.
  4. Interpretation: how strong is the evidence of a bubble in the broad market?
- **Report**: three statistics with their critical values, a list of episodes, one sentence
- Notebook: section B1

Back Exercise list

B1: solution [Solved]



- ADF -1.087 (cv_{95} -0.132); SADF 1.804 (cv_{95} 1.481); GSADF 2.194 (cv_{90} 2.028, cv_{95} 2.276); $w_0 = 58$ weeks, $L = 7$
- 5 episodes, 228 weeks in total, from 16 June 1995 to 11 February 2000; the longest: 18 April 1997 to 28 August 1998 (72 weeks). Nasdaq 100: SADF 2.722, GSADF 2.925; longest episode 23 October 1998 to 3 November 2000 (107 weeks)
- Interpretation: weaker than for the Nasdaq: SADF rejects at 5%, GSADF only at 10%, so the dated episodes are a description, not a 5% test result; the explosive phase of the S&P 500 ends before the peak of March 2000, that of the Nasdaq goes on to November 2000

TSA_ch13_seminar

B2: LPPL fits of Bitcoin in 2021 [Proposed]

- **Question**: would an LPPL fit have announced the Bitcoin peak of April 2021, and how does the estimated critical time move as the window grows?
- **Inputs**: daily Bitcoin log price from the low of 8 September 2020 (10132 USD) to the peak of 13 April 2021 (63503 USD); model: A4 and Chapter 13
- **Tasks**
  1. Fit the LPPL model (`fit_window`) on windows that start at the low and end 90, 60 and 30 days before the peak.
  2. For each fit report t_c, m, ω and the distance between t_c and the actual peak.
  3. Apply the filter (`lppl_conditions`) and say which conditions fail.
  4. Interpretation: is t_c a forecast of the crash date?
- **Report**: three critical times, six parameters, three lists of failed conditions, one sentence
- Notebook: section B2

C1: audit an AI answer [Proposed]

- A student asked an AI assistant about bubble tests and LPPL models. The answer:
- (a) To test for a bubble, run the ADF test and reject the unit root when the statistic is below -2.86.
- (b) The SADF of the Nasdaq 100 is above its critical value, so the Nasdaq was in a bubble from 1990 to 2004.
- (c) GSADF can be smaller than SADF when the bubble is at the end of the sample.
- (d) The LPPL critical time tc is the date of the crash: sell the day before tc.
- (e) A fit with m = 0.999 and omega = 1 is a valid bubble signal if its errors are small.
- (f) In a rational bubble the expected bubble grows at the rate r, although it grows faster than r while it survives.
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (Parts A and B, notebook section C1).
- **Report**: a list of six verdicts with one line of justification each

Key takeaways

- Bubble tests are right-tailed: $H_1 : \rho > 1$; a whole-sample ADF misses a bubble that collapses inside the sample
- SADF and GSADF answer "is there an episode?"; BSADF with the minimum duration $[\ln T]$ answers "when?"
- The S&P 500 shows weaker evidence than the Nasdaq 100 in the same years
- LPPL: check the filter before reading t_c; t_c moves with the window and is not a crash date
- A rational bubble grows by $(1 + r)/\pi$ while it survives and by $1 + r$ on average

After the seminar

- Chapter 13 (self-study) develops each topic of today
  - the episodes, the PSY tests, the LPPL calibration, the confidence indicator and its evaluation
- Try the [Proposed] tasks in the notebook
- A project idea: date-stamp the BET and the Bucharest sector indices with PSY and compare with the LPPL confidence indicator
- Reading: Phillips, Shi and Yu (2015a); Johansen, Ledoit and Sornette (2000); Filimonov and Sornette (2013); Shu and Zhu (2020)

References

- Blanchard, O. J., & Watson, M. W. (1982). Bubbles, rational expectations and financial markets. NBER Working Paper 945. doi:10.3386/w0945
- Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431. doi:10.1080/01621459.1979.10482531
- Filimonov, V., & Sornette, D. (2013). A stable and robust calibration scheme of the log-periodic power law model. *Physica A*, 392(17), 3698–3707. doi:10.1016/j.physa.2013.04.012
- Johansen, A., Ledoit, O., & Sornette, D. (2000). Crashes as critical points. *International Journal of Theoretical and Applied Finance*, 3(2), 219–255. doi:10.1142/S0219024900000115
- Phillips, P. C. B., Shi, S., & Yu, J. (2015a). Testing for multiple bubbles: Historical episodes of exuberance and collapse in the S&P 500. *International Economic Review*, 56(4), 1043–1078. doi:10.1111/iere.12132
- Phillips, P. C. B., Shi, S., & Yu, J. (2015b). Testing for multiple bubbles: Limit theory of real-time detectors. *International Economic Review*, 56(4), 1079–1134. doi:10.1111/iere.12131
- Phillips, P. C. B., Wu, Y., & Yu, J. (2011). Explosive behavior in the 1990s Nasdaq: When did exuberance escalate asset values? *International Economic Review*, 52(1), 201–226. doi:10.1111/j.1468-2354.2010.00625.x
- Shu, M., & Zhu, W. (2020). Detection of Chinese stock market bubbles with LPPLS confidence indicator. *Physica A*, 557, 124892. doi:10.1016/j.physa.2020.124892
- Sornette, D. (2017). *Why Stock Markets Crash: Critical Events in Complex Financial Systems*. Princeton University Press. doi:10.23943/princeton/9780691175959.001.0001